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Security & Privacy Report

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1 Introduction

This chapter lays out the essential background information on my dissertation and briefly explains what I am doing in my dissertation.

1.1 Dissertation Description

My dissertation focuses on adapting the Decentralized Realistic Urban Traffic Generator (DesRUTGe) framework [1] to Dublin. DesRUTGe [1] is a decentralized, privacy-preserving framework for realistic and high-fidelity urban traffic simulation on a microscopic traffic simulation software called Simulation of Urban MObility (SUMO) [2].

The DesRUTGe [1] framework was originally developed by researchers from Spain and evaluated on the traffic data of Barcelona. The aim of my dissertation is to:

1. Adapt the framework to traffic data from Dublin.
2. Investigate the reproducibility and adaptability of the framework in a different urban setting to Barcelona.

On a high level, the DesRUTGe [1] framework combines local Deep Reinforcement Learning (DRL) with Decentralized Federated Learning (DFL) techniques to learn to generate time-varying and high-fidelity traffic profiles on SUMO that reliably mirror the traffic profiles in a real urban traffic network.

The following list briefly explains the steps that I took to adapt the DesRUTGe [1] framework to Dublin.

1. Collect real traffic intensity data provided by SCATS¹ from Dublin City Council ².
2. Import Dublin's traffic network from OpenStreetMap (OSM)³ into the SUMO simulator.
3. Partition Dublin into Traffic Analysis Zones (TAZs) (see Figure 1.1) using Voronoi tessellation [3] based on traffic detector locations, ensuring each zone is uniquely associated with a detector and its surrounding road network.
4. Assign a local Deep Reinforcement Learning (DRL) agent to each TAZ, responsible for modeling its local traffic dynamics.
5. Train each agent using Proximal Policy Optimization (PPO) to learn a policy that generates traffic flows matching the target 24-hour traffic intensity profile.
6. Apply Decentralized Federated Learning (DFL), where agents exchange and aggregate model parameters with neighboring zones to improve generalization while preserving data locality.

In a nutshell, my dissertation focuses on adapting the DesRUTGe [1] approach, a decentralized, privacy-preserving framework for realistic urban traffic simulation, to Dublin. A microscopic traffic simulator, SUMO, is leveraged as a closed-loop simulation environment, where deep reinforcement learning agents (PPO-based) learn to generate traffic patterns that match real-world conditions. The framework integrates decentralized federated learning, where different TAZs (detector-based zones) train local DRL models and collaborate by sharing model parameters rather than raw data, improving scalability, robustness, and privacy.

¹Sydney Coordinated Adaptive Traffic System

²<https://data.smartdublin.ie/dataset/analyze/fc61ace0-1a54-483a-8f29-affe6d70a04e>

³<https://www.openstreetmap.org/>

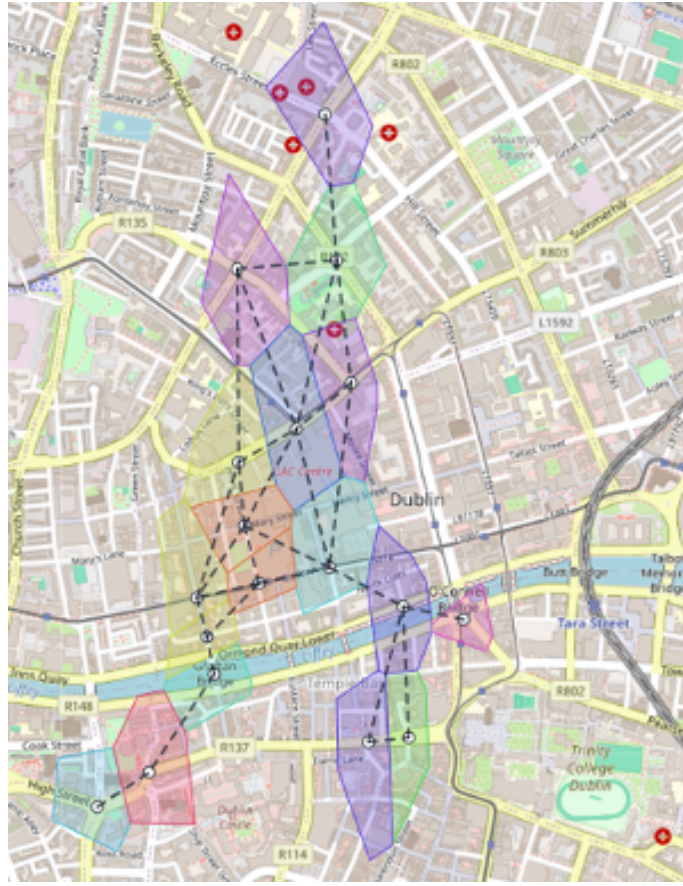


Figure 1.1: Traffic Analysis Zones from Voronoi partitioning on a selected region in Dublin City Center

2 Privacy

Traffic data is very sensitive in nature and can be used for behavior profiling, tracking and spying purposes. Moreover, distributed learning systems like this can be exposed to Byzantine attacks which involve malicious agents that try to manipulate the training process. This chapter explains the different security & privacy concerns surrounding the DesRUTGe [1] framework and how they were tackled in the dissertation. Additionally, this chapter also acknowledges the unaddressed security concerns in the current implementation of DesRUTGe [1] framework in the dissertation.

2.1 Privacy by Design

Sensitive Traffic Data Collection - GDPR

The traffic data in my dissertation corresponds to the vehicle counts recorded by real inductive loop detectors present within the road surface approaching an intersection. This data is sourced from the SCATS system installed in Dublin for adaptive signal control, and is publically available for research under the Dublinked Scheme by the Dublin City Council [4].

In order to tackle the privacy concerns surrounding the data collection process of my dissertation, the following GDPR principles were up-held.

1. **Data Anonymisation:** The SCATS system in Dublin provides traffic flow data, not personal data, since the detectors are responsible to produce vehicle count figures, and is not linked to any individual.

Additionally, the data on Dublin City Council's open portal is aggregated and non-personal.

2. **Purpose Limitation:** The sole purpose of the collected traffic intensity data in Dublin was to train and evaluate local DRL agents to learn to generate realistic traffic patterns in a closed loop in SUMO simulator.
3. **Data Minimisation:** There is zero raw-traffic data sharing between DRL agents during the Decentralized Federated Learning (DFL) phase. The DFL architecture eliminates the need for a central server to aggregate the raw data or model weights, eliminating the single-point-of-failure problem, and allowing peer-peer model parameter sharing and aggregation for improving local models.

Addressing Byzantine Attacks

Byzantine attacks occur in distributed systems when some participants act maliciously or unpredictably, sending false or conflicting information. These attacks make it difficult for the system to reach to a valid consensus, even if most participants are honest.

These attacks are relevant to the DFL architecture of the DesRUTGe [1] framework, where multiple DRL agents perform peer-to-peer model weight sharing and aggregation to update their respective models and can degrade the accuracy of the resulting models. Some examples of the types of Byzantine attacks addressed in the dissertation are:

- **Label Flipping:** A malicious client corrupts training by changing the labels of local data samples to incorrect classes before model updates are computed.
- **Noise Injection:** A malicious client perturbs its model parameters or gradients by adding random noise so that the shared update degrades global training.
- **Sign Flipping:** A malicious client reverses the sign of its model update, pushing aggregation in the opposite direction of honest learning.
- **IPM:** Inner Product Manipulation. A malicious client crafts an update from the negative scaled average of benign updates so that the aggregated model is steered away from the correct optimization direction.
- **ALIE:** A Little Is Enough. A malicious client builds a statistically deceptive update using the mean and standard deviation of observed benign updates to remain plausible while still poisoning aggregation.

In order to tackle the different types of possible Byzantine attacks, the following defense mechanisms are applied within the DesRUTGe [1] framework.

- **Mean:** The aggregated model is computed as the average of all received updates. This acts as a simple baseline aggregation method without Byzantine robustness.
- **Median:** The aggregated model is obtained by taking the coordinate-wise median of the received updates. This reduces the influence of extreme malicious values.
- **Trimmed-Mean:** The aggregation removes the largest and smallest values in each coordinate before averaging, making the result more resistant to outlier updates.
- **Krum:** The defense selects the single update that is closest to the majority of other updates to ignore anomalous malicious contributions.
- **Multi-Krum:** The defense extends Krum by selecting several updates that are mutually close and averaging them, improving robustness while using more information.
- **Clustering:** The received updates are separated into groups according to their similarity, and the largest cluster is used for aggregation under the assumption that honest clients form the dominant group.

- **WFAgg-D:** Weighted Filter Aggregation. The defense selects updates whose Euclidean distance to the coordinate-wise median is smallest, filtering out updates that are far from the benign majority.
- **WFAgg-C:** The defense selects updates whose cosine-distance behavior is closest to the median trend, reducing the impact of directionally anomalous malicious updates.
- **WFAgg-T:** The defense uses temporal statistics from previous rounds to keep updates whose distance and cosine patterns remain consistent over time, helping detect sudden malicious deviations.
- **WFAgg-E:** The defense applies an exponential moving aggregation between the local model and neighbors' updates, smoothing the influence of shared updates across rounds.
- **Alt-WFAgg:** The defense combines Multi-Krum, clustering, and temporal filtering into a weighted scheme so that only updates supported by several criteria strongly influence aggregation.
- **WFAgg:** The defense combines all previous distance-based, cosine-based, and temporal filtering into a weighted aggregation rule that emphasizes updates judged reliable by multiple robustness tests.

Having mentioned the different Byzantine attacks and the supported defense mechanisms, it is important to mention that the framework isn't a fully Byzantine-secure federated learning system because currently, it lacks authenticated identities, secure transport, trust management, and input validation. Additionally, several of the above mentioned defenses rely on knowing or tuning the expected malicious fraction before the simulation starts.

2.2 Potential Security Risks

This section goes over the unaddressed, but ever-present security risks and threats that currently concern the implementation of DesRUTGe [1] in my dissertation.

1. The biggest and most obvious security risk in the current implementation is that the DRL agents exchange model updates in the DFL phase over plain TCP and deserialize them with pickle. There is no encryption or authentication during communication between agents and the received data (model parameters) is directly unpickled. This creates a risk of exposure to remote code execution, spoofed agent identities, replayed updates, and man-in-the-middle tampering.
2. Another issue is privacy leakage through model sharing in the DFL phase. Although the raw data is not shared between agents, the full model parameters are shared without any differential privacy (introducing noise to the data), secure aggregation, clipping, or protection against gradient inversion. A malicious agent could potentially infer sensitive information about local training data from the model updates.
3. There could be a potential Denial of Service (DoS) attack since in the current implementation, the agents wait for expected updates whenever they finish their respective reinforcement learning round. Although it is normal for different agents to have different training times depending on their location and target traffic intensities, an agent that withholds messages or floods the queue could create denial-of-service behavior.

3 Conclusion

This report has outlined the key security and privacy considerations concerning the adaptation of the DesRUTGe framework to a Dublin urban traffic context. The privacy-by-design principles discussed in Chapter 2 demonstrate that the framework handles sensitive traffic data responsibly and respects the GDPR principles of data anonymisation, purpose limitation, and data minimisation by relying on publicly available,

aggregated SCATS data from Dublin City Council. The decentralized federated learning architecture further reinforces these principles by eliminating the need for a central aggregation server, ensuring raw traffic data remains local to each Traffic Analysis Zone at all times and eliminating the single-point-of-failure problem.

Furthermore, while the framework supports a range of robust aggregation defenses against Byzantine attacks, it currently lacks authenticated identities, secure transport, and input validation, meaning it does not fully support Byzantine-secure federated learning. The unaddressed security risks discussed in Section 2.2, particularly the use of unencrypted pickle-based communication, the absence of differential privacy protections, and the potential for denial-of-service behavior, represent important limitations of the current implementation that would need to be resolved before any real-world deployment of the framework.

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